

Internship Management System — Database Systems Project

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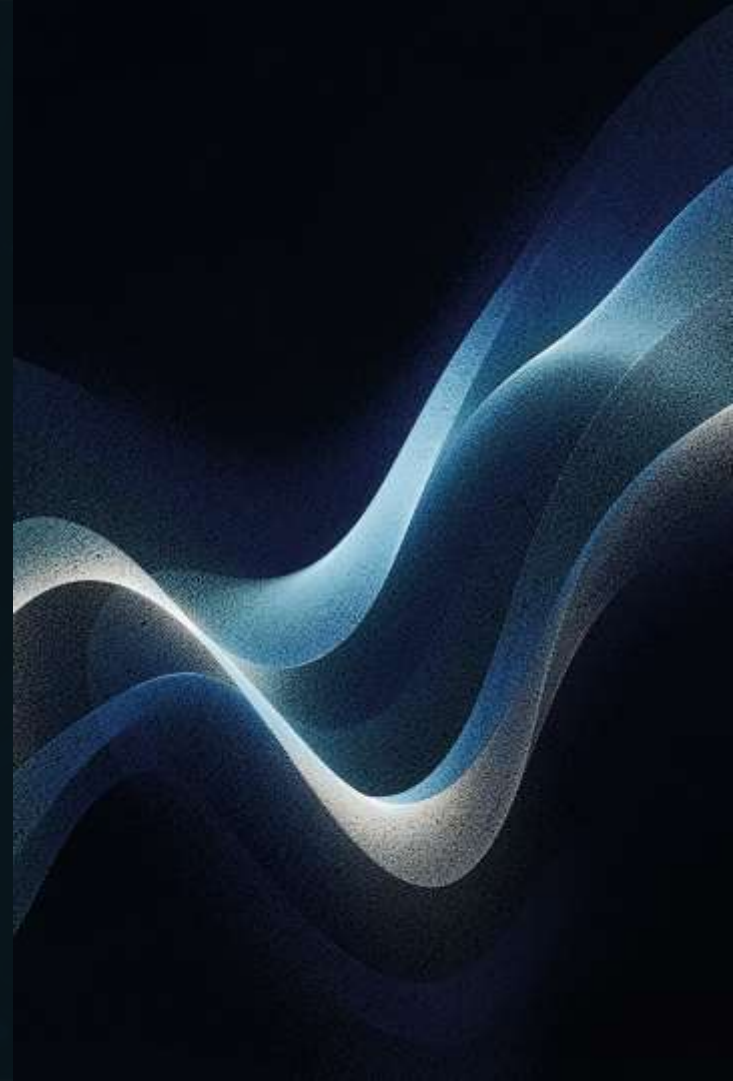
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Problem Statement

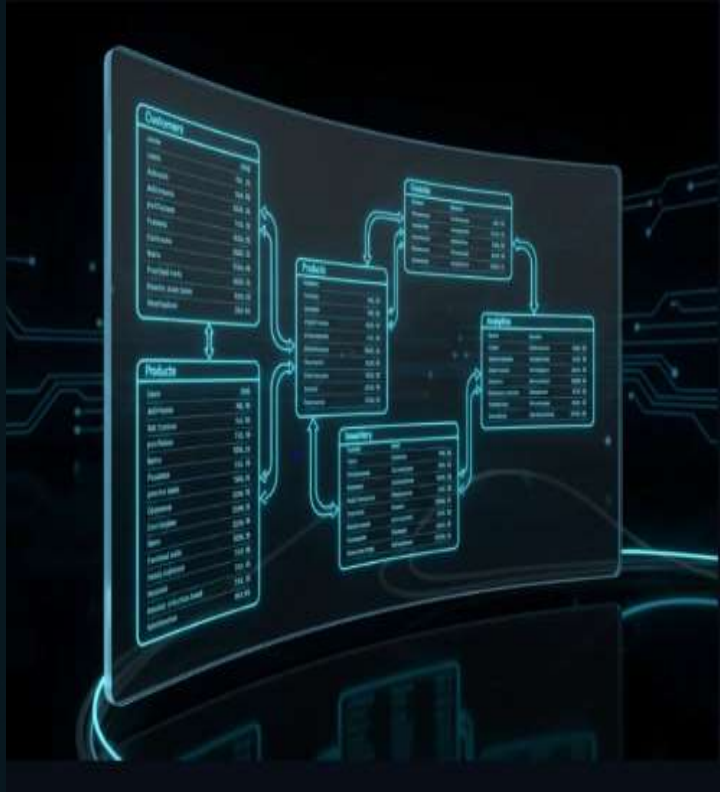
Universities rely on fragmented manual processes for internship administration: paper records, scattered emails, and spreadsheets. This causes poor tracking, delayed approvals, limited transparency, and higher administrative overhead.



An abstract graphic on the left side of the slide featuring several overlapping, wavy bands of blue and teal colors against a dark background, creating a sense of movement and depth.

Solution Snapshot

A web-based Internship Management System that centralizes listings, applications, progress tracking, evaluations, and credit verification — connecting students, employers, and coordinators in one secure portal.



Project Objectives

- Digitize internship opportunities and applications
- Connect students, employers, and coordinators
- Track progress, weekly reports, and evaluations
- Automate viva assessments and credit verification
- Maintain secure, auditable records with role-based access

SCHEMA

USERS

ID	name	email	password	role	created_at
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COORDINATOR_PROFILES

ID	user_id	department
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EMPLOYERS

ID	user_id	company_name	industry
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APPLICATIONS

ID	listing_id	student_id	cover_letter	status
dept_comments		created_at		

STUDENTS

ID	user_id	department	credit_hours	status
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INTERNSHIP_LISTINGS

ID	employer_id	title	description	positions	status
target_department		created_at			

INTERNSHIPS

ID	student_id	employer_id	listing_id	status	start_date	end_date
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PROGRESS_REPORTS

ID	internship_id	student_id	report_text	file_url	approved
submitted_at					

EVALUATIONS (Score)

ID	internship_id	employer_score	viva_score
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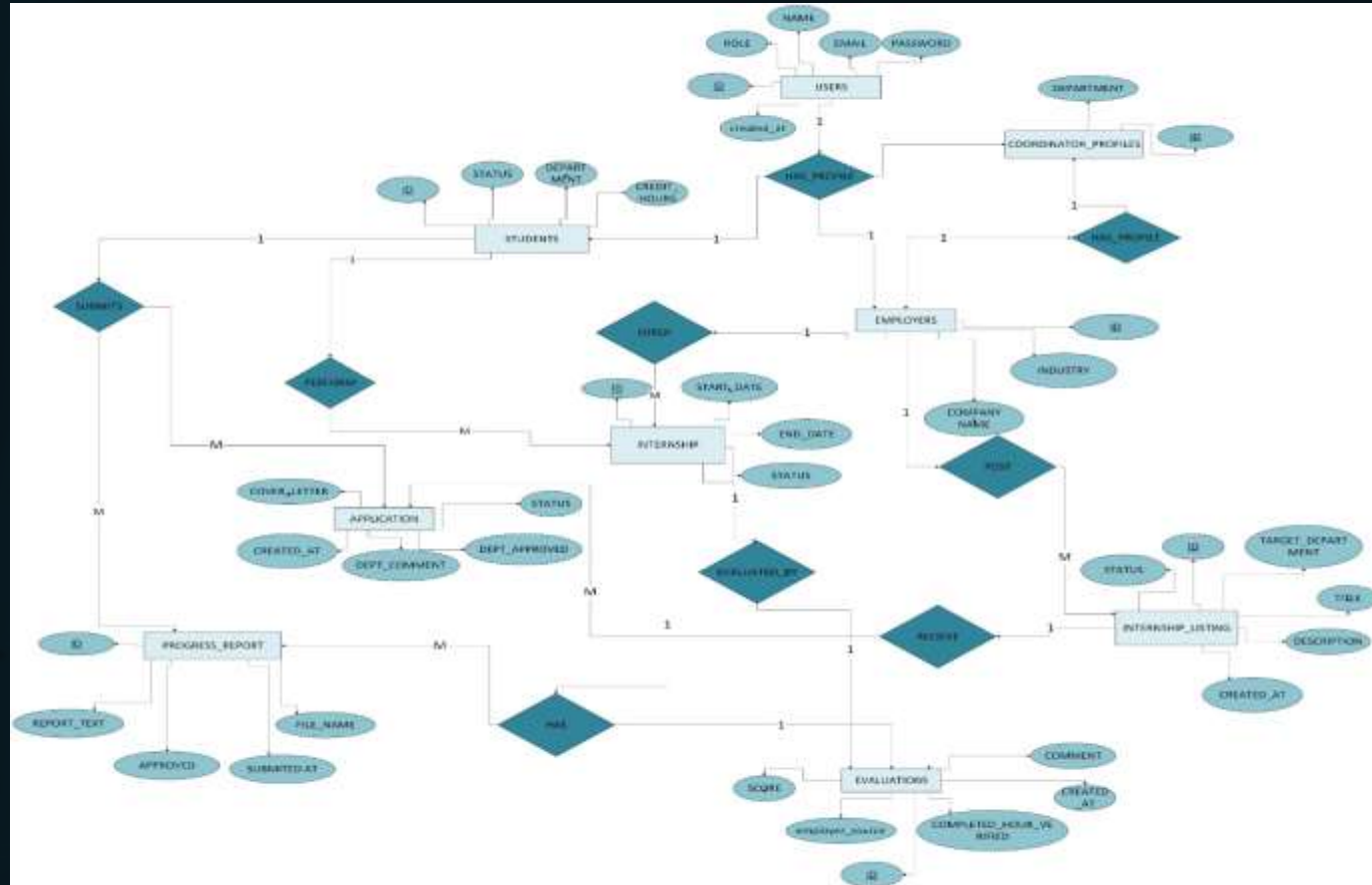
EVALUATIONS (Verification)

ID	coordinator_id	comments	completed_hours_verified
updated_at			

VIVA_SUBMISSIONS

ID	viva_id	student_id	answers	file_url	submitted_at
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ENTITY RELATIONSHIP DIAGRAM



Technology Stack



Frontend

HTML • CSS • JavaScript • Bootstrap — responsive UI and accessible forms



Backend

Flask (Python) — RESTful API endpoints, business logic, and session handling



Database

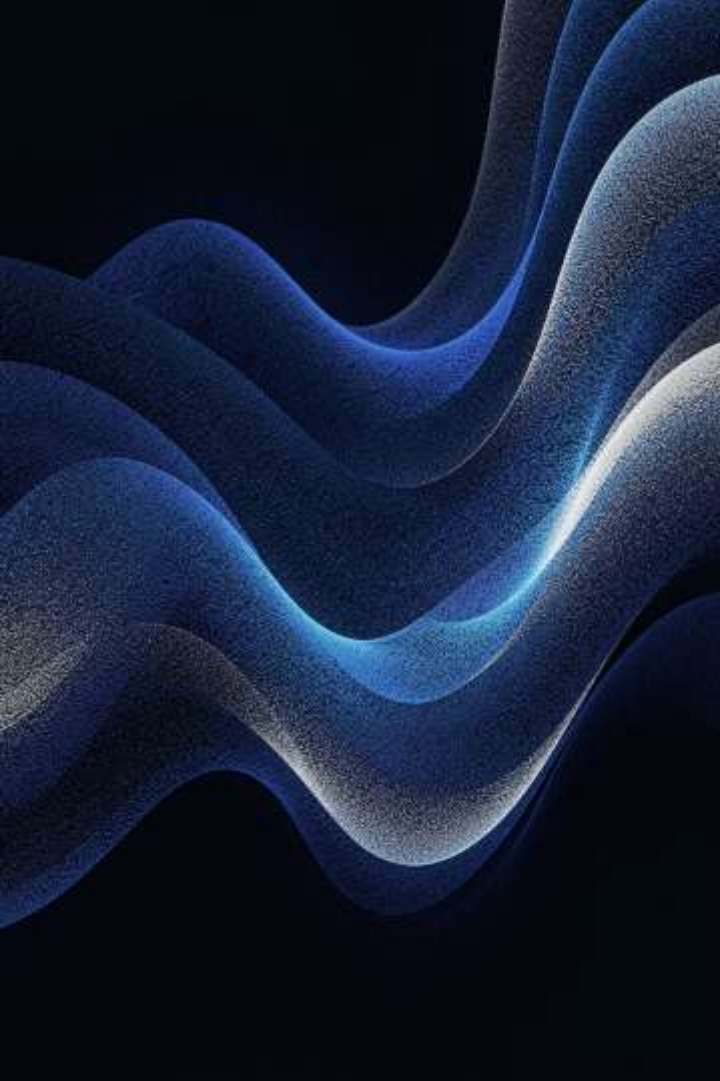
PostgreSQL / Supabase — relational storage, constraints, and secure access



Authentication

JWT tokens — stateless authentication and role-based authorization





System Architecture — Three Tier

Frontend UI → Flask REST API → PostgreSQL Database

Scalability

Clear separations allow independent scaling and caching.

Security

API layer enforces authentication, DB stores encrypted sensitive data.

Maintainability

Modular services simplify testing and feature updates.



Database Design

Core tables and relationships: Users (students, employers, coordinators), InternshipListings, Applications, Internships, Evaluations, VivaAssessments, WeeklyReports, Workflow, CreditVerification. Keys, constraints, and junction tables model many-to-many relationships (e.g., students $\leftrightarrow$ listings).



User Management Module

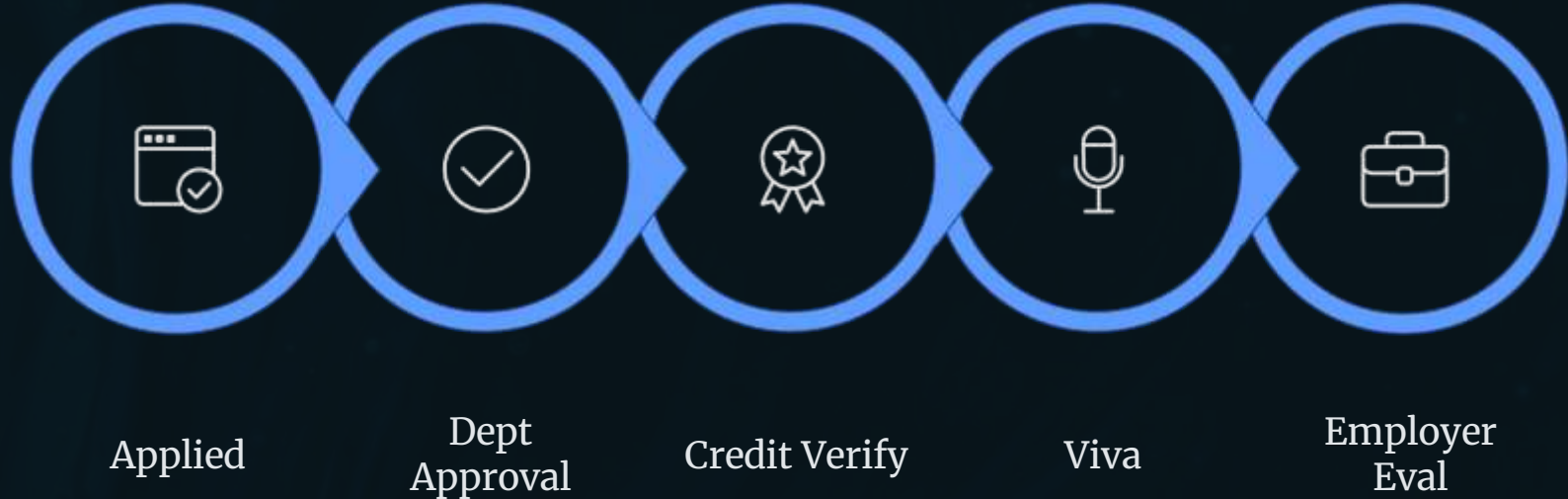
- Registration and secure login flows
- JWT authentication with token refresh and expiry
- Role-Based Access Control (Student, Employer, Coordinator)
- Profile management and audit logging for traceability



Internship & Application Module

- Employers create and manage listings with deadlines and requirements
- Students browse, filter, and submit applications with attachments
- Application lifecycle tracking (Submitted → Reviewed → Approved/Rejected)
- Automated notifications and document validation

Workflow, Assessment & Reporting



Key modules supported: – Workflow engine tracking stage transitions and timestamps – Progress reports: weekly submissions, file uploads, coordinator approvals – Evaluation & viva: structured rubrics, score storage, feedback comments – Credit verification: coordinator validations and automated credit assignment

Sign In Interface



The image shows a sign-in interface for an 'Internship System'. It features a white card with rounded corners on a light gray background. The card has a title 'Internship System' in blue, followed by the subtitle 'Sign in to your account' in gray. There are two input fields: 'Email' with the placeholder 'Enter your email' and 'Password' with the placeholder 'Enter your password'. Below these is a blue 'Sign In' button. At the bottom, there is a link 'Don't have an account? Register here'.

Internship System

Sign in to your account

Email

Enter your email

Password

Enter your password

Sign In

Don't have an account? [Register here](#)

Registration Interface

Create Account

Full Name

Email

Password

Role

-- Select your role --

Create Account

Already have an account? [Sign in](#)

Student Dashboard

Internship Management System

fizzah [Logout](#)

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Active

Overview

Browse Internships

My Applications

Progress Reports

Viva Assessment

1

Applications

1

Internships

0

Evaluations

0

Viva Done

Internship Progress

✓

Application

Submitted & Approved

2

Credit Verification

Pending HOD Review

3

Viva Assessment

Not scheduled

4

Employer Evaluation

Pending

Coordinator Dashboard

IMS — Coordinator

mulethaComputer ScienceLogout

OVERVIEW

Dashboard

WORKFLOW MANAGEMENT

Dept Approval

All Workflows

Internship Progress

ACADEMIC

Students

Visa Monitor

Credit Hours

Dashboard Overview

Monitor all internship activity across the department.

1

PENDING APPROVALS

0

ACTIVE WORKFLOWS

No active workflows yet.

Go to Dept Approval

Employer Dashboard

